

Electrical Measurement Analytics

DROP TABLE IF EXISTS electrical_analysis;

```
CREATE TABLE electrical_analysis(  
  "Date" DATE,  
  "Time" TIME,  
  "Substation" TEXT,  
  "Transformer" TEXT,  
  "Voltage (kV)" NUMERIC(10,2),  
  "Current (A)" NUMERIC(10,2),  
  "Frequency (Hz)" NUMERIC(10,2),  
  "Power Factor" NUMERIC(10,2),  
  "Active Power (MW)" NUMERIC(10,2),  
  "Reactive Power (MVAR)" NUMERIC(10,2),  
  "Apparent Power (MVA)" NUMERIC(10,2),  
  "Energy (MWh)" NUMERIC(10,2),  
  "Temperature (°C)" NUMERIC(10,2),  
  "Humidity (%)" NUMERIC(10,2),  
  "Load (%)" NUMERIC(10,2),  
  "Fault Status" VARCHAR(100)  
);
```

SELECT * FROM electrical_analysis

COPY electrical_analysis

FROM 'D:\\VV.csv'

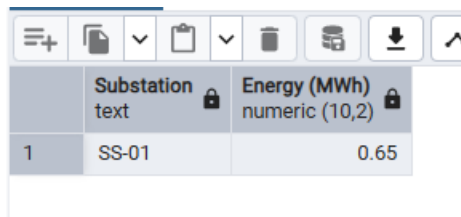
DELIMITER ';

CSV HEADER;

-- Load Analysis

-- 1.Which substation has the highest annual energy consumption?

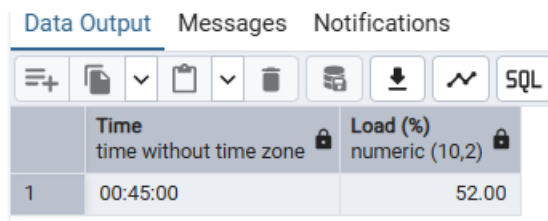
```
SELECT "Substation","Energy (MWh)" FROM electrical_analysis ORDER BY "Energy (MWh)"  
DESC LIMIT 1;
```



	Substation text	Energy (MWh) numeric (10,2)
1	SS-01	0.65

-- 2.What are the peak load hours?

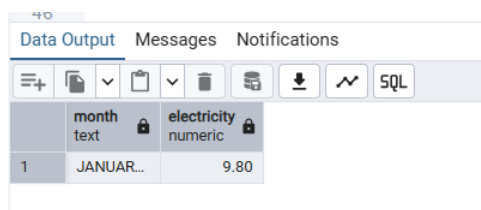
```
SELECT "Time","Load (%)" FROM electrical_analysis ORDER BY "Time" DESC LIMIT 1;
```



	Time time without time zone	Load (%) numeric (10,2)
1	00:45:00	52.00

-- 3.Which month records the highest electricity demand?

```
SELECT TO_CHAR( "Date",'MONTH')AS MONTH,SUM("Active Power (MW" )  
AS ELECTRICITY FROM electrical_analysis GROUP BY "Date" ORDER BY ELECTRICITY  
DESC LIMIT 1;
```



	month text	electricity numeric
1	JANUAR...	9.80

-- Power Quality

-- Which transformers have power factor below 0.90?

```
SELECT * FROM electrical_analysis
```

```
SELECT "Transformer","Power Factor" FROM electrical_analysis WHERE "Power Factor"< 0.90;
```

Data Output Messages Notifications		
SQL		
	Transformer text	Power Factor numeric (10,2)
1	TR-101	0.96
2	TR-101	0.95
3	TR-101	0.94
4	TR-101	0.94

-- Is voltage maintained within $\pm 5\%$ of the nominal value?

```
SELECT ROUND(AVG("Voltage (kV)",2), ROUND(AVG("Voltage (kV)")*1.05,2) AS  
UPPER_LIMIT,  
ROUND(AVG("Voltage (kV)")*.95,2) AS LOWER_LIMIT  
FROM electrical_analysis
```

Data Output Messages Notifications			
SQL			
	round numeric	upper_limit numeric	lower_limit numeric
1	11.18	11.74	10.62

```
SELECT "Voltage (kV)" ,CASE WHEN "Voltage (kV)" BETWEEN 10.45 AND 11.55 THEN 'WITH  
IN  $\pm 5\%$ '
```

```
ELSE 'OUT_OF_RANGE' END AS NOMINAL_VALUE
```

```
FROM electrical_analysis
```

Data Output Messages Notifications		
	Voltage (kV) numeric (10,2)	nominal_value text
1	11.21	WITH IN +-5%
2	11.19	WITH IN +-5%
3	11.17	WITH IN +-5%
4	11.15	WITH IN +-5%

-- Which substations experience frequent frequency deviations?

```

WITH CTE AS(
SELECT "Substation",ROUND(AVG("Frequency (Hz)"),2) AS FF FROM electrical_analysis
GROUP BY "Substation"
)
SELECT E."Substation",(E."Frequency (Hz)"-C.FF) AS FREDEV FROM CTE C JOIN
electrical_analysis E ON
E."Substation"=C."Substation" ORDER BY FREDEV DESC LIMIT 1;

```

Data Output Messages Notifications		
	Substation text	fredev numeric
1	SS-01	0.02

-- Which transformers operate above 50% load most often?

```

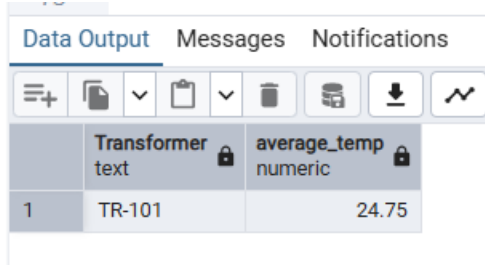
SELECT "Transformer","Load (%)" FROM electrical_analysis WHERE "Load (%)" > 50;

```

Data Output Messages Notifications		
	Transformer text	Load (%) numeric (10,2)
1	TR-101	48.00
2	TR-101	49.00
3	TR-101	50.00
4	TR-101	52.00

-- Which transformer has the highest average temperature?

```
SELECT "Transformer",ROUND(AVG("Temperature (°C)"),2) AS AVERAGE_TEMP FROM  
electrical_analysis  
GROUP BY "Transformer" ORDER BY AVERAGE_TEMP DESC LIMIT 1;
```



	Transformer text	average_temp numeric
1	TR-101	24.75

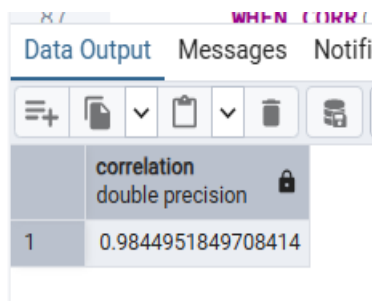
-- Which transformer shows the highest fault frequency?

```
SELECT "Transformer",COUNT(*) AS fault_count FROM electrical_analysis WHERE "Fault  
Status" <> 'Normal' GROUP BY "Transformer"  
ORDER BY fault_count DESC LIMIT 1;
```

Blank

-- Does ambient temperature affect electrical load?

```
SELECT * FROM electrical_analysis  
SELECT CORR("Temperature (°C)", "Load (%)") AS correlation FROM electrical_analysis;
```



	correlation double precision	
1	0.9844951849708414	

ver factor?

```
SELECT CORR("Humidity (%)", "Power Factor") AS correlation
FROM electrical_analysis;
```

	correlation double precision
1	0.8528028654224421

-- How does temperature affect transformer loading?

```
SELECT CASE
  WHEN CORR("Temperature (°C)", "Load (%)") > 0 THEN 'Positive Relationship'
  WHEN CORR("Temperature (°C)", "Load (%)") < 0 THEN 'Inverse (Negative) Relationship'
  ELSE 'No Relationship' END AS relationship FROM electrical_analysis;
```

	relationship text
1	Positive Relations...

-- Which substation has the highest outage duration?

```
SELECT "Time","Fault Status"FROM electrical_analysis WHERE "Fault Status"<>'Normal';
```

Blank

-- Which fault type occurs most frequently?

```
SELECT "Fault Status"FROM electrical_analysis WHERE "Fault Status"<>'Normal';
```

-- Which transformer requires preventive maintenance?

Blank

```
SELECT "Transformer", COUNT(*) AS maintenance_alerts FROM electrical_analysis
WHERE "Load (%)" > 80 OR "Temperature (°C)" > 35 OR "Power Factor" < 0.90 OR "Fault
Status" <> 'Normal'
GROUP BY "Transformer"
ORDER BY maintenance_alerts DESC
```

Blank